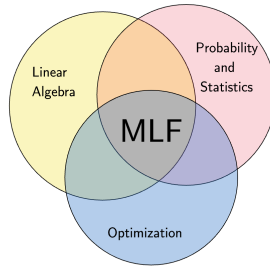


Week-6



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Disclaimer: These notes were generated as a part of this [live session](#). It can't be used as a standalone resource or a set of notes for this week.

1. SVD

Singular value decomposition is a way of decomposing a matrix into three different matrices. This can also be called matrix factorization. We have already seen one such decomposition before: eigen-decomposition. For diagonalizable matrices:

$$A = PDP^{-1}$$

For symmetric matrices this becomes

$$A = QDQ^T$$

where, Q is orthogonal ($Q^T Q = QQ^T = I$).

What is the SVD?

SVD Theorem (real version).

Any matrix A of shape $m \times n$ can be represented as:

$$A = U\Sigma V^T$$

where

- U is an orthogonal matrix of shape $m \times m$
- Σ is a diagonal-like matrix of shape $m \times n$
- V is an orthogonal matrix of shape $n \times n$

Every matrix has an SVD. Recall that only square matrices, that too only diagonalizable square matrices, have an eigen-decomposition.

How do we compute U, Σ, V ? To derive the SVD, we will have to understand some properties of the closely related symmetric matrices $A^T A$ and AA^T .

2. $A^T A$ and AA^T

Let A be an $m \times n$ matrix. We will study the matrices AA^T and $A^T A$. This will help us derive the SVD of A . Let us look at the shapes of these matrices:

- $A^T A$ is an $n \times n$ matrix.
- AA^T is an $m \times m$ matrix.

For detailed proofs of these properties, refer to this [document](#).

2.1. Property-1

$A^T A$ and AA^T are symmetric.

Proof

$$(A^T A)^T = A^T A$$

$$(AA^T)^T = AA^T$$

2.2. Property-2

$A^T A$ and AA^T have non-negative eigenvalues.

Proof: Let (λ, v) be some eigenvalue-eigenvector pair for $A^T A$, $v \neq 0$.

$$\begin{aligned}(A^T A)v &= \lambda v \\ \lambda v &= (A^T A)v \\ \text{multiply with } v^T & \quad \lambda(v^T v) = v^T (A^T A)v \\ &= v^T A^T A v \\ &= (Av)^T (Av) \\ &= \|Av\|^2\end{aligned}$$

Now $v^T v$ is positive, $\|Av\|^2$ is also non-negative. Therefore, $\lambda \geq 0$. Also note that $A^T A$ and AA^T are positive semi-definite matrices.

2.3. Property-3

$A^T A$ and AA^T have the same *positive* eigenvalues.

Proof: Let (λ, v) be some eigenvalue-eigenvector pair for $A^T A$, $v \neq 0$ such that $\lambda > 0$.

$$(A^T A)v = \lambda v$$

To bring AA^T into the picture, left-multiply by A :

$$(AA^T)(Av) = \lambda(Av)$$

Av is an eigenvector of AA^T with eigenvalue λ provided $Av \neq 0$. [show why $Av \neq 0$]. What we have shown is this: every positive eigenvalue of $A^T A$ is an eigenvalue of AA^T . We also need to show that every positive eigenvalue of AA^T is also an eigenvalue of $A^T A$.

2.4. Property-4

The rank of the matrices A , $A^T A$ and AA^T are equal.

$$\text{rank}(A^T A) = \text{rank}(AA^T) = \text{rank}(A)$$

2.5. Property-5

$A^T A$ and AA^T have r positive eigenvalues, where $r = \text{rank}(A)$. Also:

- $A^T A$ will have $n - r$ zero eigenvalues
- AA^T will have $m - r$ zero eigenvalues

For example, if A has rank 4 and is of shape 10×7

- $A^T A$ will have 3 zero eigenvalues
- AA^T will have 6 zero eigenvalues

3. SVD: Derivation

We will work with either $A^T A$ or AA^T depending on which of m, n is smaller. For this section, we will work with $A^T A$. Recall that we are looking at $A = U\Sigma V^T$.

3.1. Building V

Since $A^T A$ is symmetric, by the spectral theorem, $\mathbb{R}^n$ will have a basis of orthonormal vectors that are eigenvectors of $A^T A$. Also $A^T A$ has r positive eigenvalues and the rest are zero. Arranging these in descending order:

- $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r \geq \lambda_{r+1} = \dots = \lambda_n = 0$
- $\{v_1, \dots, v_n\}$ is an orthonormal basis for $\mathbb{R}^n$.
 - $\|v_i\| = 1$
 - $v_i^T v_j = 0$, for $i \neq j$

Let's put all these n vectors into the columns of a matrix V and all the n eigenvalues into a matrix D . V is orthogonal.

$$V = \begin{bmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{bmatrix}, D = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$$

$$A^T A = V D V^T$$

3.2. Building Σ

Notice that $\lambda_i \geq 0$ for $1 \leq i \leq n$. We are going to define:

$$\sigma_i^2 = \lambda_i$$

or

$$\sigma_i := \sqrt{\lambda_i}$$

σ_i is called the i^{th} singular value of A . Note that:

$$\bullet \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$$

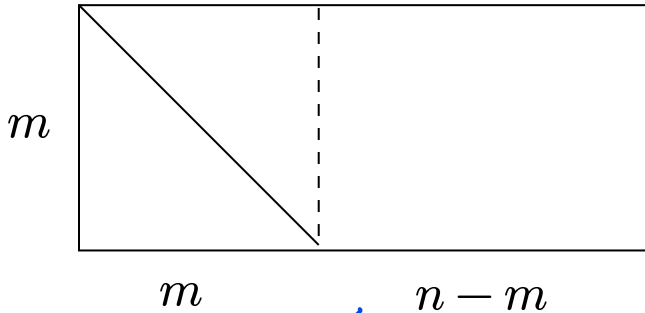
$$\bullet \sigma_{r+1} = \cdots = \sigma_{\min(m,n)} = 0$$

The singular values occupy the diagonal positions of Σ . It is a diagonal-like matrix:

$$\Sigma = \begin{bmatrix} \sigma_1 & & & & \\ & \sigma_2 & & & \\ & & \ddots & & \\ & & & \sigma_r & \\ & & & & \sigma_{r+1} \\ & & & & & \ddots \\ & & & & & & \sigma_{\min(m,n)} \end{bmatrix}$$

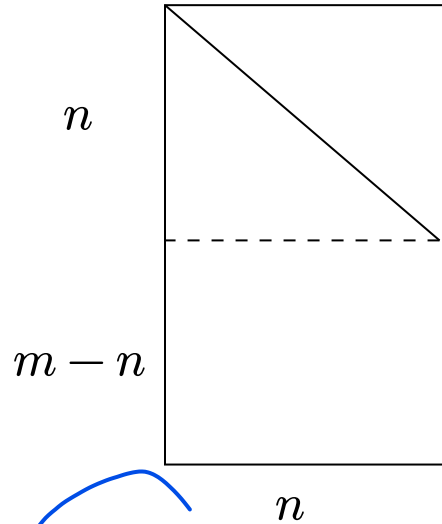
If you have an $m \times n$ matrix with rank r , it has r positive singular values and $\min(m, n) - r$ singular values that are zero.

Wide matrix $m < n$



$\min(m, n) = m$
 m singular values

Tall matrix $m > n$



$\min(m, n) = n$
 n singular values

3.3. Build U

We want to construct the following orthogonal matrix for $m \times m$:

$$U = \begin{bmatrix} | & & | \\ u_1 & \cdots & u_m \\ | & & | \end{bmatrix}$$

Consider the following assignment for $1 \leq i \leq r$:

$$u_i = \frac{Av_i}{\sigma_i}$$

This is a valid operation as A is $m \times n$ and $v_i \in \mathbb{R}^n$, so $Av_i \in \mathbb{R}^m$, and $\sigma_i > 0$ for $1 \leq i \leq r$. We shall now check if $\{u_1, \dots, u_r\}$ are orthonormal.

$$\begin{aligned}
 u_i^T u_j &= \frac{(Av_i)^T (Av_j)}{\sigma_i \sigma_j} \\
 &= \frac{v_i^T A^T A v_j}{\sigma_i \sigma_j} \\
 &= \frac{v_i^T (A^T A) v_j}{\sigma_i \sigma_j} = \begin{cases} 1, & i = j \\ 0 & i \neq j \end{cases} \\
 &= \frac{v_i^T (\lambda_j v_j)}{\sigma_i \sigma_j} \\
 &= \frac{\sigma_i \sigma_j}{\lambda_j (v_i^T v_j)} \\
 &= \frac{\sigma_i \sigma_j}{\sigma_i \sigma_j}
 \end{aligned}$$

What we can now do is extend $\{u_1, \dots, u_r\}$ to $\{u_1, \dots, u_m\}$ which is an orthonormal basis for $\mathbb{R}^m$.

Remark: The derivation is incomplete. A detailed derivation will be added as notes for those interested at a later point.

Rearranging $u_i = \frac{Av_i}{\sigma_i}$, we get an important equation for $1 \leq i \leq r$.

$$Av_i = \sigma_i u_i$$

Some important terminology that we need to get used to:

- u_i is called a left-singular vector of A
- v_i is called a right singular vector of A

4. Examples

$$A = \begin{bmatrix} 3 & 2 \\ -2 & 0 \end{bmatrix}$$

Step-1: V

Step-1.1

$$A^T A = \begin{bmatrix} 3 & -2 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} 13 & 6 \\ 6 & 4 \end{bmatrix}$$

Step-1.2

$$\begin{aligned} |A^T A - \lambda I| &= \begin{vmatrix} 13 - \lambda & 6 \\ 6 & 4 - \lambda \end{vmatrix} \\ &= (13 - \lambda)(4 - \lambda) - 36 \\ &= \lambda^2 - 17\lambda + 16 \\ &= (\lambda - 16)(\lambda - 1) \end{aligned}$$

The eigenvalues are $\lambda = 16, 1$.

Step-1.3

$$\begin{aligned} (A^T A - 16I)v &= 0 \\ \begin{bmatrix} -3 & 6 \\ 6 & -12 \end{bmatrix} v &= 0 \\ \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} &= 0 \end{aligned}$$

$\frac{1}{\sqrt{5}}(2, 1)$ is a unit norm eigenvector for $\lambda = 16$.

$$\begin{aligned} (A^T A - I)v &= 0 \\ \begin{bmatrix} 12 & 6 \\ 6 & 3 \end{bmatrix} v &= 0 \\ \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} &= 0 \end{aligned}$$

$\frac{1}{\sqrt{5}}(1, -2)$ is a unit norm eigenvector for $\lambda = 1$.

$$V = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$$

Step-2: Σ

$$\sigma_1 = \sqrt{\lambda_1} = 4$$

$$\sigma_2 = \sqrt{\lambda_2} = 1$$

$$\Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$$

Step-3: U

Using

$$u_i = \frac{Av_i}{\sigma_i}$$

We have:

$$u_1 = \frac{1}{4} \cdot \frac{1}{\sqrt{5}} \begin{bmatrix} 3 & 2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$= \frac{1}{4} \cdot \frac{4}{\sqrt{5}} \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$u_2 = \frac{1}{1} \cdot \frac{1}{\sqrt{5}} \cdot \begin{bmatrix} 3 & 2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$= \frac{1}{\sqrt{5}} \begin{bmatrix} -1 \\ -2 \end{bmatrix}$$

We now have:

$$U = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ -1 & -2 \end{bmatrix}.$$

Finally:

$$\begin{bmatrix} 3 & 2 \\ -2 & 0 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$$

Example-2

$$A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 1 \\ 3 & 2 & 1 \end{bmatrix}$$

Step-1: V

Step-1.1

$$\begin{aligned} A^T A &= \begin{bmatrix} 1 & -1 & 3 \\ 2 & 0 & 2 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 1 \\ 3 & 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 11 & 8 & 5 \\ 8 & 8 & 8 \\ 5 & 8 & 11 \end{bmatrix} \end{aligned}$$

Step-1.2

$$|A^T A - \lambda I| = \begin{vmatrix} 11 - \lambda & 8 & 5 \\ 8 & 8 - \lambda & 8 \\ 5 & 8 & 11 - \lambda \end{vmatrix}$$

The eigenvalues are $\lambda = 24, 6, 0$. Note that the rank of $A^T A$ is 2.

Step-1.3

Solving for $(A^T A - \lambda I)v = 0$ will give you:

$$V = \frac{1}{\sqrt{6}} \begin{bmatrix} \sqrt{2} & -\sqrt{3} & -1 \\ \sqrt{2} & 0 & 2 \\ \sqrt{2} & \sqrt{3} & -1 \end{bmatrix}$$

Step-2:

$$\Sigma = \begin{bmatrix} 2\sqrt{6} & 0 & 0 \\ 0 & \sqrt{6} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Step-3:

$$u_1 = \frac{Av_1}{\sigma_1} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$u_2 = \frac{Av_2}{\sigma_2} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

Since u_3 is orthogonal to u_1, u_2 and has norm 1, we can find u_3 . How?

If you assume $u_3 = (a, b, c)$:

$$\begin{aligned} a + c &= 0 \\ a + b - c &= 0 \end{aligned} \implies a = 1, b = -2, c = -1$$

Now normalize to get $u_3 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}$ ✓

$$U = \frac{\sqrt{6}}{\sqrt{6}} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \\ 0 & 1/\sqrt{3} & -2/\sqrt{6} \\ 1/\sqrt{2} & -1/\sqrt{3} & -1/\sqrt{6} \end{bmatrix}$$

$$\Rightarrow U = \frac{1}{\sqrt{6}} \begin{bmatrix} \sqrt{3} & \sqrt{2} & 1 \\ 0 & \sqrt{2} & -2 \\ \sqrt{3} & -\sqrt{2} & -1 \end{bmatrix}$$

5. Properties of SVD

Recall that the SVD of $A = U\Sigma V^T$, where:

$$U = \begin{bmatrix} | & & | \\ u_1 & \cdots & u_m \\ | & & | \end{bmatrix}, V = \begin{bmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{bmatrix}$$

$$\Sigma = \text{diag-like}(\sigma_1, \cdots, \sigma_{\min(m,n)})$$

5.1. Property-1

We can express A as the following sum of r rank-1 matrices, where $r = \text{rank}(A)$.

$$A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \cdots + \sigma_r u_r v_r^T$$

Each of $u_i v_i^T$ is a rank-1 matrix of dimension $m \times n$.

5.2. Property-2

The four fundamental subspaces of A can be read off from the left and right singular vectors:

- column space: $\mathcal{C}(A) = \text{span}\{u_1, \dots, u_r\}$
 - The first r left-singular vectors span the column space
 - The column space is a subspace of $\mathbb{R}^m$
- left nullspace: $\mathcal{N}(A^T) = \text{span}\{u_{r+1}, \dots, u_m\}$
 - The last $m - r$ left-singular vectors span the left nullspace
 - The left nullspace is a subspace of $\mathbb{R}^m$

$$\text{span}\{u_1, \dots, u_r\} \oplus \text{span}\{u_{r+1}, \dots, u_m\} = \mathbb{R}^m$$

- row space: $\mathcal{C}(A^T) = \text{span}\{v_1, \dots, v_r\}$
 - The first r right-singular vectors span the row space
 - The row space is a subspace of $\mathbb{R}^n$
- nullspace: $\mathcal{N}(A) = \text{span}\{v_{r+1}, \dots, v_n\}$
 - The last $n - r$ right-singular vectors span the nullspace
 - The nullspace is a subspace of $\mathbb{R}^n$

$$\text{span}\{v_1, \dots, v_r\} \oplus \text{span}\{v_{r+1}, \dots, v_n\} = \mathbb{R}^n$$

Proof Outline

We have:

$$A = \sigma_1 u_1 v_1^T + \dots + \sigma_r u_r v_r^T$$

Now, Av_i is:

$$\begin{aligned} Av_i &= (\sigma_1 u_1 v_1^T + \dots + \sigma_r u_r v_r^T) v_i \\ &= \sigma_1 u_1 v_1^T v_i + \dots + \sigma_r u_r v_r^T v_i \\ &= \sigma_i u_i v_i^T v_i \\ &= \sigma_i u_i \end{aligned}$$

We also have:

$$A^T = \sigma_1 v_1 u_1^T + \dots + \sigma_r v_r u_r^T$$

$$A^T u_i = \sigma_i v_i$$

We can make use of these equations to prove the result.

Singular value equations

The following are very important equations:

$$\begin{aligned} Av_i &= \sigma_i u_i \\ A^T u_i &= \sigma_i v_i \end{aligned}, \quad 1 \leq i \leq r$$

The left and right singular vectors are connected with A and A^T via the singular value. This should be reminiscent of the eigenvector equation for square matrices.

5.3. Property-3

- The left singular vectors, namely the columns of U , are the normalized eigenvectors of AA^T .
- The right singular vectors, namely the columns of V , are the normalized eigenvectors of $A^T A$.
- The square of the non-zero singular values are the non-zero eigenvalues of $A^T A$ and AA^T .

6. Another Example

$$A = \begin{bmatrix} 1 & 1 & 1 & -1 \\ 0 & 1 & 1 & 0 \\ -1 & 1 & 1 & 1 \end{bmatrix}$$

- A is of shape 3×4
- AA^T

Step-1: AA^T

$$AA^T = \begin{bmatrix} 4 & 2 & 0 \\ 2 & 2 & 2 \\ 0 & 2 & 4 \end{bmatrix}$$

Step-2: U

$$U = \frac{1}{\sqrt{6}} \begin{bmatrix} \sqrt{2} & -\sqrt{3} & -1 \\ \sqrt{2} & 0 & 2 \\ \sqrt{2} & \sqrt{3} & -1 \end{bmatrix}$$

Step-3: Σ

$$\Sigma = \begin{bmatrix} \sqrt{6} & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Step-4: V

$$v_i = \frac{A^T u_i}{\sigma_i}, \quad 1 \leq i \leq 2$$

$$v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \quad v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

By eyeballing, we can get v_3 and v_4 :

$$V = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

7. Applications

7.1. Low-Rank Approximation

If the SVD of A is:

$$A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots + \sigma_r u_r v_r^T$$

then we can approximate A with the help of a matrix with a lower rank. Some of the places where this is used:

- Recommendation engines
- Image compression
- PCA (dimensionality reduction)

The "best" matrix of rank $k < r$ that approximates A is given by:

$$\hat{A} = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots + \sigma_k u_k v_k^T$$

$\hat{A}$ is as close to A as possible, meaning, the approximation error is very low. One immediate application is image compression. Every image is a matrix. For example, an image could be a 1000×1000 matrix. Each cell is called a pixel. These pixels usually range from $[0, 255]$. A black and white image is just a matrix.

7.2. Pseudo-inverse

If A is a matrix of shape $m \times n$, then its pseudo-inverse is a matrix of shape $n \times m$ that has inverse-like properties and is computed using the SVD. It exists for every matrix and it is equal to the inverse for invertible matrices.

If the SVD of A is:

$$A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots + \sigma_r u_r v_r^T$$

Then the **Moore-Penrose** inverse:

A dagger

$$A^\dagger = \frac{1}{\sigma_1} v_1 u_1^T + \dots + \frac{1}{\sigma_r} v_r u_r^T$$

If you have a system of equations $Ax = b$, the least square solution to $Ax = b$ is given by:

$$\hat{x} = A^\dagger b$$